Result 1 (saved graded map). [Topology foundations, graded map right identity across local scopes] Let K be a type. Let *field* be a field structure on K. Let C be a chain complex of modules over K in natural degrees. Let $M = H_*(C)$. Let f be a degree-preserving map from M to M. Let *clean_bound_0* and *clean_comparison_identity_0* be natural numbers. Let $N = H_*(C)$. Let $g = (\mathrm{id}_N \circ f)$. Then

$$g = f.$$

Result 2 (saved relative graded map). [Topology foundations, relative graded map right identity across local scopes]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Let P be two nested subsets of X with inclusion evidence. Let $M = H_*(P; H)$. Let f be a degree-preserving map from M to M. Let *clean_bound_0*, *clean_pi_bound_0* and *clean_comparison_identity_0* be natural numbers. Let $N = H_*(P; H)$. Let $g = (\mathrm{id}_N \circ f)$. Then

$$g = f.$$

LLM-assisted example and proof (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.